AN INVERSE LIMIT OF NONEMPTY QUASI-COMPACT HAUSDORFF SPACES IS NONEMPTY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the limit as the intersection of the closed subsets cut out by the graphs of the transition maps inside the product. Invoked at the start of the counted proof (source lines 2732-2740). Presents the limit as an intersection of closed subsets of a quasi-compact space.
- The statement that an intersection of closed subsets of a quasi-compact space is nonempty when every finite subfamily has nonempty intersection. Invoked inside the counted proof (source lines 2737-2739). Reduces nonemptiness of the limit to the finite intersection property.

Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-inverse-limit-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 2692–2692. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered category and let $i \mapsto X_i$ be a diagram over $\mathcal{I}$ in the category of topological spaces. If each X_i is quasi-compact, Hausdorff, and nonempty, then* $\lim X_i$ *is nonempty.*

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Proof. In the proof of Lemma lemma-inverse-limit-quasi-compact we have seen that $X = \lim X_i$ is the intersection of the closed subsets

$$Z_\varphi = \Gamma_\varphi \times \prod_{l \neq j,k} X_l$$

inside the quasi-compact space $\prod X_i$ where $\varphi : j \to k$ is a morphism of $\mathcal{I}$ and $\Gamma_\varphi \subset X_j \times X_k$ is the graph of the corresponding morphism $X_j \to X_k$. Hence by Lemma lemma-intersection-closed-in-quasi-compact it suffices to show any finite intersection of these subsets is nonempty. Assume $\varphi_t : j_t \to k_t$, $t = 1, \ldots, n$ is a finite collection of morphisms of $\mathcal{I}$. Since $\mathcal{I}$ is cofiltered, we can pick an object j and a morphism $\psi_t : j \to j_t$ for each t. For each pair t, t' such that either (a) $j_t = j_{t'}$, or (b) $j_t = k_{t'}$, or (c) $k_t = k_{t'}$ we obtain two morphisms $j \to l$ with $l = j_t$ in case (a), (b) or $l = k_t$ in case (c). Because $\mathcal{I}$ is cofiltered and since there are finitely many pairs (t, t') we may choose a map $j' \to j$ which equalizes these two morphisms for all such pairs (t, t'). Pick an element $x \in X_{j'}$ and for each t let x_{j_t}, resp. x_{k_t} be the image of x under the morphism $X_{j'} \to X_j \to X_{j_t}$, resp. $X_{j'} \to X_j \to X_{j_t} \to X_{k_t}$. For any index $l \in \mathrm{Ob}(\mathcal{I})$ which is not equal to j_t or k_t for some t we pick an arbitrary element $x_l \in X_l$ (using the axiom of choice). Then $(x_i)_{i \in \mathrm{Ob}(\mathcal{I})}$ is in the intersection

$$Z_{\varphi_1} \cap \ldots \cap Z_{\varphi_n}$$

by construction and the proof is complete. $\square$